

Understanding OpenStack Terms

These OpenStack concepts are the core building blocks for provisioning virtual clusters.

- **Project:** This acts as a boundary for team isolation and quotas. Think of it as the fenced-in plot of land your team is assigned; it keeps your resources completely separate from other groups and sets a hard limit on how much compute power you are allowed to spin up. In your course, this is typically owned by the instructor or OpenStack itself.
- **Image:** An OS template. This is essentially your raw, unconfigured installation file—like a base Linux image—that the cloud provider makes available for you to boot your virtual machines from.
- **Flavor:** This dictates the vCPU and RAM shape of a virtual machine. It is exactly like picking hardware specs from a catalog; a "small" flavor might give you 2 CPUs and 4GB of RAM, while a "large" flavor provisions 8 CPUs and 16GB of RAM. The provider defines these shapes.
- **Network:** A virtual L2 network. Imagine this as the physical Ethernet switch that connects all your nodes together in a local, isolated loop. You will define this infrastructure using Terraform.
- **Subnet:** This provides the CIDR block, gateway, and DHCP on a network. If the Network is the physical switch, the Subnet is the logic handing out local IP addresses (like `192.168.x.x`) to your nodes so they actually know how to route data to one another. Terraform manages this.
- **Router:** This connects your private networks to the provider network. It acts as the gateway or doorway out of your isolated cluster environment, allowing your nodes to reach the wider campus network or the internet. It is owned by Terraform.
- **Port:** A virtual NIC (Network Interface Card). It is the specific digital jack on a virtual machine where your virtual network cable plugs in. This is managed by Terraform or OpenStack's compute service, Nova.
- **Security group:** A cloud firewall enforced before the guest OS. Even if your custom Linux setup has misconfigured local `iptables`, this acts as a bouncer at the very perimeter of the instance, dropping unauthorized traffic before it ever touches the operating system. You control these rules via Terraform.
- **Floating IP:** A provider-routable IP mapped to a private port. By default, your cluster nodes might only have private, internal IP addresses. Attaching a Floating IP gives a

specific node a public address so that external users can reach it directly over the network. Terraform handles this mapping.

- **Cinder volume:** Persistent block storage. Think of this as a virtual external hard drive plugged into a node. If a node crashes or gets deleted, the data safely survives on the Cinder volume. Depending on the layer, it is owned by Terraform or CSI.